

FORE SCHOOL OF MANAGEMENT, NEW DELHI

PGDM-BDA | Term 04 | Academic Year 2026–27

AGENTIC AI FOR BUSINESS AUTOMATION

PROJECT REPORT

AI Business Growth Advisor

A Domain-Expert Multi-Agent System for Sales & Marketing Decision Support

Project Detail Information	
Course	Agentic AI for Business Automation
Program	PGDM-BDA Term 04 Academic Year 2026–27
Institution	FORE School of Management, New Delhi
Domain	FMCG / Retail Sales & Marketing Decision Support
Architecture	Orchestrator + specialised analytical agents
AI Runtime	Ollama — llama3.2:3b
Data	Synthetic / demonstration FMCG business datasets
Interface	Streamlit

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Table of Contents

Section	Page
Abstract	iii
1. Introduction	1
2. Business Need and Problem Analysis	1
3. Gap Analysis	2
4. Objectives and Scope	2
5. Course Alignment and Deliverables	3
6. System Architecture and Workflow	4
7. Agent and Component Design	5
8. Technology Stack	7
9. Data Design	7
10. Implementation Methodology	8
11. Workflow and Data Processing	9
12. Dashboard and Outputs	10
13. Results and Discussion	12
14. Limitations and Risks	13
15. Ethical, Privacy and Governance Considerations	13
16. Evaluation Mapping	14
17. Future Scope	14
18. Conclusion	15
19. How to Run, References and Project Artefacts	15

Abstract

This project develops an AI-assisted, domain-expert multi-agent decision-support system for FMCG and retail sales and marketing analysis. The application helps managers move from raw business data to evidence-based decisions by combining structured data analysis, customer and outlet analysis, market intelligence, strategy prioritisation and management reporting in one workflow.

A Streamlit interface allows a user to select a business location and year or upload a new business dataset. An Orchestrator Agent coordinates specialised agents: the Data Analyst Agent evaluates revenue, units, availability, SKU and category performance; the Customer / Outlet Agent identifies outlet priorities; the Market Intelligence Agent evaluates competitor signals; the Strategy Agent converts evidence into prioritised actions; and the Report Agent generates an executive report and 30-day action plan.

The prototype runs locally using Ollama and llama3.2:3b, with Python, Pandas, NumPy, scikit-learn and Streamlit. The demonstration dataset contains 30,000 sales observations and linked product, outlet, promotion and competitor datasets. The Gurgaon 2026 demonstration identifies a Q2-to-Q3 revenue decline of 26.92%, a 15.15% decline in units and a 2.30 percentage-point decline in availability.

The system is a decision-support prototype rather than an autonomous commercial execution system. Its strategy layer explicitly distinguishes evidence from causal claims and treats competitor movements and availability changes as signals unless causality is established.

1. Introduction

FMCG and retail managers work with multiple dimensions of business information simultaneously. Revenue can change because of SKU mix, availability, outlet performance, promotions, pricing or competitive conditions. A dashboard can display these measures, but management still has to interpret them and decide what deserves attention first.

The AI Business Growth Advisor addresses this gap by combining analytical components with specialised AI agents. Rather than asking one general-purpose model to perform every task, the system separates responsibilities into domain-oriented roles. The Orchestrator coordinates the agents, while each specialist produces evidence that contributes to the final recommendation.

The application is built around the management questions: What is happening? Where is it happening? What evidence supports the diagnosis? What should management investigate or do next?

2. Business Need and Problem Analysis

The core problem is not the absence of data; it is the gap between available data and timely decision-making. A manager may have sales files, SKU information and competitor observations but still spend substantial time manually joining files, comparing periods and determining which issues deserve priority.

- Performance visibility across revenue, units, availability and trends.
- Identification of products, outlets and segments that require attention.
- Market context alongside internal performance.
- Prioritisation of actions rather than an unstructured list of metrics.
- Repeatable analysis across cities, years and uploaded datasets.
- Governed recommendations that distinguish evidence from unsupported causal claims.

3. Gap Analysis

Gap	Conventional Approach	Project Response
G1 — Fragmented analysis	Different files and metrics are analysed separately.	Orchestrator coordinates specialised analytical agents.
G2 — Descriptive rather than prescriptive	Dashboards show what happened but may not prioritise actions.	Strategy Agent converts evidence into ranked actions.
G3 — Outlet blind spot	Overall city revenue can hide outlet-level deterioration.	Customer / Outlet Agent identifies priority outlet types and outlets.
G4 — Weak market context	Internal sales may be viewed without competitor signals.	Market Intelligence Agent evaluates competitor price and availability.
G5 — Fixed demonstration	A hard-coded city limits reuse.	Upload mode supports new business datasets and dynamic city/year selection.
G6 — Unsupported causal claims	Associations can be described as causes.	Strategy rules distinguish evidence, signals and causality.

4. Objectives and Scope

- Build an AI-powered FMCG/Retail business decision-support workflow.
- Use specialised agents for sales, customer/outlet and market analysis.
- Coordinate the agents through an Orchestrator Agent.
- Prioritise evidence-linked actions through a Strategy Agent.
- Generate a management-ready report and 30-day action plan.
- Provide a Streamlit interface for city/year analysis and uploaded business data.
- Maintain a clear distinction between evidence, inference and causality.

In scope: sales, SKU, category, outlet, promotion, competitor, quarterly and Q2-to-Q3 analysis; recommendations; dashboard presentation; dataset upload and validation.

Out of scope: automatic price changes, distributor orders, live CRM execution, financial transactions and autonomous commercial execution.

5. Course Alignment and Deliverables

Course Requirement	Project Implementation
Agentic workflow	Orchestrator coordinates specialised agents and passes evidence through the decision pipeline.
Specialised agents	Data Analyst, Customer/Outlet, Market Intelligence, Strategy and Report agents.
Local AI	Ollama with llama3.2:3b for local inference.
Business workflow	Data → Diagnosis → Customer/Outlet → Market → Strategy → Report.
Evidence-driven decisions	Deterministic Python analysis supplies exact numerical evidence.
Interactive interface	Streamlit dashboard with city/year selection and upload mode.
Reusable input	CSV upload supports sales, product, outlet, promotion and competitor datasets.
Demonstration	Gurgaon 2026 FMCG/Retail dataset with 30,000 sales observations.

6. System Architecture and Workflow

The architecture separates user interaction, orchestration, specialised analysis and management reporting. The Orchestrator coordinates the specialists and passes their outputs to the Strategy and Report layers.

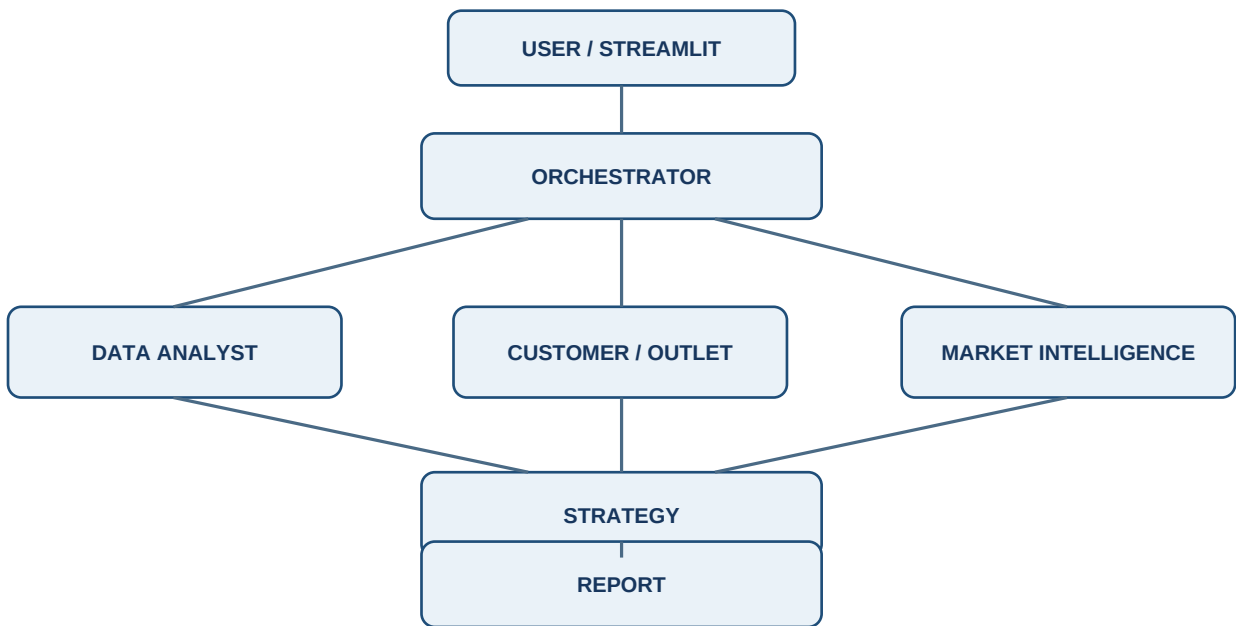


Figure 1: Multi-agent architecture of the AI Business Growth Advisor

At runtime, the user starts from Streamlit. The Orchestrator calls the analytical components, consolidates their outputs, sends the evidence to the Strategy Agent and finally passes the strategy to the Report Agent.

7. Agent and Component Design

Component	Responsibility	Typical Output
Orchestrator Agent	Coordinates the end-to-end analysis.	Integrated analysis and final report.
Data Analyst Agent	Calculates KPIs, trends, SKU/category performance and period comparisons.	Numerical evidence.
Customer / Outlet Agent	Analyses outlet types, tiers and individual outlet changes.	Outlet priorities and insight.
Market Intelligence Agent	Analyses competitor price, availability and market signals.	Market summary.
Strategy Agent	Prioritises actions and produces diagnosis.	Top actions and 30-day plan.
Report Agent	Structures findings into a management report.	Executive report.

7.1 Data Analyst Agent

The Data Analyst Agent is the quantitative foundation. It loads linked datasets, cleans column names, parses dates and calculates city, SKU, category, monthly, quarterly, promotion and competitor-related metrics. It performs Q2-to-Q3 comparisons at overall and SKU level.

7.2 Customer / Outlet Agent

This agent focuses on where performance changes occur. It compares outlet types and individual outlets across periods and calculates contribution shares. In the demonstration, Kirana contributes the largest revenue share while Eating & Dining records the largest outlet-type decline.

7.3 Market Intelligence Agent

This component evaluates competitor prices, availability, brands, categories and quarterly movement. Its outputs are treated as market signals rather than automatic causal explanations.

7.4 Strategy Agent

The Strategy Agent is a hybrid decision layer. Python performs exact numerical prioritisation while the local LLM provides concise executive interpretation. Guardrails prevent unsupported claims about competitor pricing or availability.

7.5 Report Agent

The Report Agent converts strategy and analysis into a consistent management report containing business performance, priorities, market intelligence, actions and a 30-day plan.

7.6 Orchestrator Agent

The Orchestrator initialises the five major components and executes the analysis sequence. The successful end-to-end run confirms that the backend components can work together.

8. Technology Stack

Technology	Purpose
Python 3.14	Core implementation and orchestration.
Pandas	Data loading, joins, grouping and business analysis.
NumPy	Numerical operations.
scikit-learn	Analytical/ML utilities and future extensibility.
Ollama + llama3.2:3b	Local language-model inference.
Streamlit	Interactive dashboard and upload interface.
python-dotenv	Configuration support.
Plotly	Interactive visualisation capability.
CSV	Primary data input format.

9. Data Design

The demonstration uses five connected FMCG/Retail datasets. The sales table contains 30,000 observations and provides the central transaction-level analytical grain.

Dataset	Key Fields	Role
Sales	Date, Region, City, Outlet_ID, SKU_ID, Units_Sold, Revenue, Selling_Price, Promotion_ID, Availability_Percent, Competitor_Price	Core performance analysis.
Product	SKU_ID, Product_Name, Category, Subcategory, Pack_Size_ml, MRP, Margin_Percent, Channel, Launch_Date	SKU/category enrichment.
Outlet	Outlet_ID, Outlet_Name, City, Region, Outlet_Type, Tier, Monthly_Footfall, Distributor_ID, Active	Outlet segmentation.
Promotion	Promotion_ID, Promotion_Type, SKU_ID, Start_Date, End_Date, Discount_Percent, POSM_Type, Target_Channel, Campaign_Budget	Promotion analysis.
Competitor	Date, Competitor, Competitor_SKU, Category, Pack_Size_ml, Price, Promotion, City, Availability_Percent	Market intelligence.

The demonstration contains intentional business scenarios including a Gurgaon Q3 availability shock for selected SKUs, competitor price/availability pressure, promotion fields, outlet variation and seasonal/regional variation.

10. Implementation Methodology

10.1 Scaffolding and data

The project was organised into agents, data, outputs, prompts and tools directories, with app.py as the Streamlit entry point. The backend was built and tested before the UI was connected.

10.2 Analytical agents

The Data Analyst and Customer/Outlet components use deterministic calculations. The Market Intelligence component similarly produces structured summaries, ensuring that core numbers are not dependent on probabilistic generation.

10.3 Local LLM integration

Ollama was installed locally and llama3.2:3b was tested through Python. The LLM is used where language interpretation is useful, while exact KPI computation remains in Python.

10.4 Strategy and reporting

The Strategy Agent combines exact evidence with controlled LLM interpretation. The Report Agent structures the resulting strategy into a management-facing output.

10.5 Streamlit and upload mode

The Streamlit application provides a demo-data mode and an upload mode. Sales, product and outlet CSV files are required for uploaded analysis, while promotion and competitor data are optional supporting inputs. Required columns are validated and cities/years are detected from uploaded sales data.

10.6 Verification

The Orchestrator was executed successfully through all major analytical steps. The Streamlit application then completed an end-to-end Gurgaon analysis with the expected Q2-to-Q3 values.

10.7 Hardening

Input validation, robust KPI extraction, explicit strategy guardrails and a clear evidence-versus-causality distinction were added to improve reliability.

11. Workflow and Data Processing

Step	Input	Processing	Result
1. Data selection	Demo data or uploads	Select source and validate files.	Ready dataset.
2. City/year	Cities and dates	Detect available values.	Analysis scope.
3. Data analysis	Linked datasets	Calculate KPIs, trends and comparisons.	Business evidence.
4. Customer/outlet	Sales + outlet	Compare outlet types and outlets.	Customer priorities.
5. Market analysis	Competitor data	Compare price and availability.	Market context.
6. Strategy	All evidence	Prioritise actions.	Recommendations.
7. Report	Strategy + analyses	Structure executive output.	Management report.
8. Dashboard	Final result	Present KPIs and tabs.	Decision interface.

The workflow keeps data processing and language generation separate. Numerical evidence is established first, after which the strategy layer interprets and prioritises it.

12. Dashboard and Outputs

The Streamlit dashboard is the user-facing layer. Demo Dataset mode provides the prepared Gurgaon 2026 analysis. Upload New Dataset mode allows new sales, product and outlet CSV files, with promotion and competitor data supported as additional inputs. Cities and years are detected from uploaded sales data.

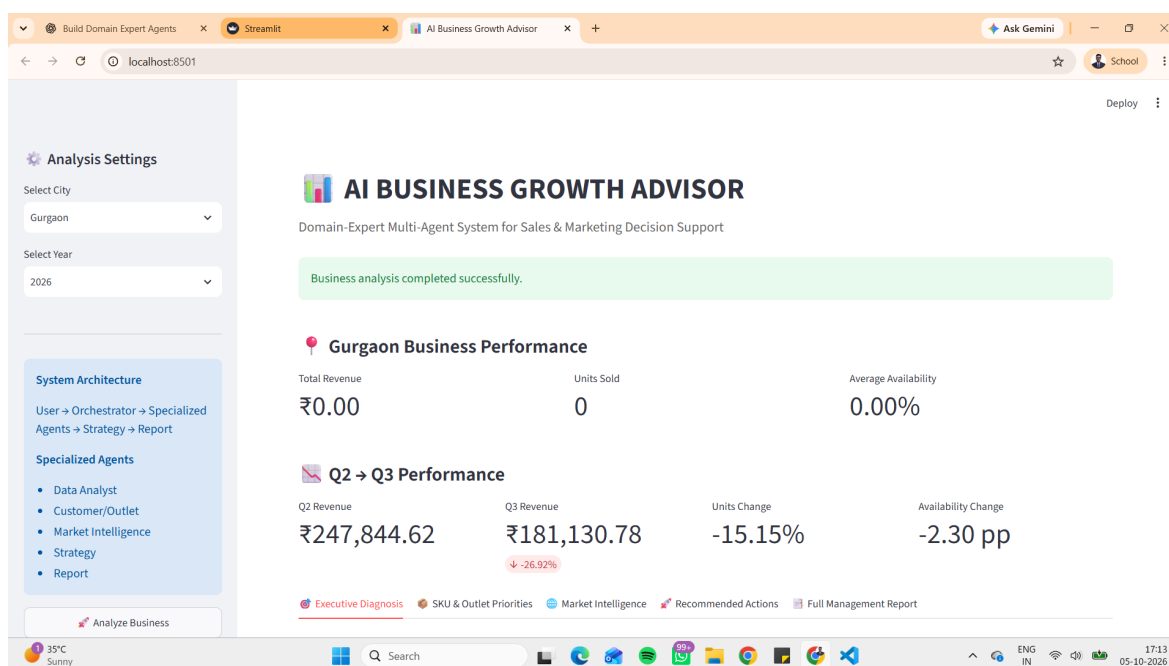


Figure 2: Streamlit dashboard showing business KPIs and Q2-to-Q3 performance.

The dashboard is organised into five tabs: Executive Diagnosis; SKU & Outlet Priorities; Market Intelligence; Recommended Actions; and Full Management Report.

Output	Purpose
Total Revenue / Units / Availability	High-level business health.
Q2 → Q3 Performance	Quarter-to-quarter movement.
Executive Diagnosis	Overall business issue.
SKU & Outlet Priorities	Products and outlets requiring attention.
Market Intelligence	Competitor metrics and signals.
Recommended Actions	Top actions and 30-day plan.

Output	Purpose
Full Management Report	Complete structured output.

13. Results and Discussion

Metric	Observed Result
Total revenue	■1,825,754.07
Units sold	49,674
Average availability	90.47%
Revenue rank	#1 among 14 observed cities
Q2 revenue	■247,844.62
Q3 revenue	■181,130.78
Q2 → Q3 revenue change	-26.92%
Q2 → Q3 units change	-15.15%
Q2 → Q3 availability change	-2.30 pp

13.1 SKU Findings

SKU005 shows a 74.02% revenue decline, 73.94% unit decline and 24.33 percentage-point availability decline. SKU004 shows a 72.86% revenue decline and 18.82 percentage-point availability decline. SKU001 shows a 52.21% revenue decline and 19.94 percentage-point availability decline. These become availability-recovery priorities.

SKU012 and SKU021 also show large revenue declines but positive availability changes, so the system classifies them as further-investigation cases rather than automatically labelling them availability problems.

13.2 Customer / Outlet Findings

Kirana contributes approximately 41.30% of city revenue. Eating & Dining records the largest outlet-type revenue decline at 47.75%, followed by Kirana at 30.87% and Convenience Store at 22.01%. Individual outlets with very large declines are also surfaced.

13.3 Market Findings

Competitor average price changes by approximately -21.83% from Q2 to Q3, while competitor availability improves by approximately 6.05 percentage points. The system reports these as signals and does not claim that competitors caused the company's decline.

13.4 Recommended Actions

- Investigate and recover SKU005 availability.
- Investigate Eating & Dining performance.
- Investigate priority high-decline outlets.
- Investigate SKU012 and other major declines where availability does not explain the movement.
- Monitor competitor pricing and availability.

14. Limitations and Risks

Limitation	Effect / Response
Synthetic / demonstration data	Results demonstrate workflow and logic rather than real-world predictive accuracy.
No causal inference	Associations are treated as signals; causality is not asserted automatically.
Local small LLM	Language quality depends on llama3.2:3b; numerical analysis remains deterministic.
CSV schema dependency	Uploaded files must follow the expected structure.
Optional data gaps	Missing promotion/competitor files limit those analyses.
No autonomous execution	Recommendations require management action.
Local upload storage	Uploaded CSVs are written into the project data path for active analysis.
FMCG/Retail focus	The present data model is domain-specific.

15. Ethical, Privacy and Governance Considerations

- Upload only business data the user is authorised to process.
- Production deployments should use access control, encryption and retention policies.
- Local Ollama inference reduces dependence on external LLM APIs in the prototype.
- Recommendations should remain reviewable by a human manager.
- Evidence, inference and uncertainty should be clearly separated.
- Uploaded files should be validated to avoid misleading outputs.
- Production versions should maintain audit logs and role-based access.

16. Evaluation Mapping

Dimension	Evidence
Functional	End-to-end analysis of sales, SKU, outlet and market data with a management report.
Agentic workflow	Orchestrator coordinates specialised agents.
Analytical depth	Monthly, quarterly, SKU, category, outlet, promotion and competitor analysis.
Decision support	Strategy Agent ranks actions and creates a 30-day plan.
Reusability	Upload mode supports new datasets and dynamic city/year detection.
Responsible AI	Guardrails distinguish signals from unsupported causal claims.
Interface	Streamlit dashboard presents KPIs, tabs and final report.

17. Future Scope

- Add interactive Plotly trend and SKU contribution charts.
- Support Excel files and downloadable data templates.
- Add stronger schema mapping for differently named columns.
- Introduce predictive models for demand, sales and availability risk.
- Add scenario analysis for price, promotion and distribution changes.
- Add RAG over company strategy documents, brand guidelines and market reports.
- Add richer customer segmentation and basket analysis.
- Add authenticated multi-user deployment with role-based access.
- Connect to ERP/CRM/distributor systems in a controlled read-only manner before automated actions.
- Add automated PDF/Excel management-report export.

18. Conclusion

The AI Business Growth Advisor demonstrates how an agentic AI architecture can be applied to a practical sales and marketing decision-support problem. The system does more than display metrics: it coordinates specialised analytical components, identifies where performance is changing, interprets market signals, prioritises management actions and produces an action plan.

The Gurgaon 2026 demonstration shows the complete journey from data to decision. A 26.92% Q2-to-Q3 revenue decline is decomposed into SKU, outlet and availability evidence, while competitor movements are retained as signals rather than unsupported causal explanations.

The upload mode further changes the application from a fixed demonstration into a reusable prototype. A user can provide a new business dataset, validate the structure, select a detected city and year, and run the same analytical pipeline.

19. How to Run, References and Project Artefacts

19.1 How to Run

- Open the project folder in VS Code.
- Ensure Ollama is installed and llama3.2:3b is available.
- Install dependencies from requirements.txt.
- Run **streamlit run app.py** from the project root.
- Open the local Streamlit URL.
- Use Demo Dataset → Gurgaon → 2026 → Analyze Business, or select Upload New Dataset and upload the supported CSV files.
- Review the executive diagnosis, SKU/outlet priorities, market intelligence, recommended actions and full management report.

19.2 Project Artefacts

Artefact	Purpose
app.py	Streamlit application and user interface.
agents/data_analyst_agent.py	Business and KPI analytics.
agents/customer_outlet_agent.py	Outlet and customer analysis.
agents/market_intelligence_agent.py	Competitor and market analysis.
agents/strategy_agent.py	Decision prioritisation and action plan.
agents/report_agent.py	Management report generation.
agents/orchestrator_agent.py	End-to-end agent coordination.

Artefact	Purpose
data/*.csv	Demonstration FMCG/Retail datasets.
requirements.txt	Python dependencies.
README.md	Project documentation.

19.3 References

1. Agentic AI for Business Automation (AAIBA), PGDM-BDA Term 04, FORE School of Management.
2. Streamlit documentation — interactive Python application framework.
3. Ollama documentation — local large-language-model runtime.
4. Pandas documentation — data analysis and manipulation library.
5. scikit-learn documentation — machine-learning and analytical utilities.

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